



- 1) Write a segment of code that remove all zero elements from a stack, leaving all other elements.  
NOTE: You do not know any knowledge of how the stack is implemented

- 2) What is the output of the following segment of code – Verify your answer with drawing

```
Stack S;  
ClearStack(&S);  
int X=1, Y=0, Z=4;  
Push(X+Z, &S);  
while(Z!=Y)  
{  
    Push(Z, &S);  
    Z--  
}  
while(!IsEmpty(S))  
{  
    Pop(&X, &S);  
    Printf("%d\n", X);  
}
```

3) Given the following specification of the Pop operation:

**void Pop(Stack \*S, StackEntry \*E, int \* UnderFlow)**

Function: Remove top element from a stack and return it.

Precondition: A stack has been created

Postcondition: If stack is empty UnderFlow variable will have the value 1

Otherwise UnderFlow variable will have the value 0, the top element of the stack will be deleted, and its value will returned using E variable

a) Write a function to implement this operation using the array-based stack implementation

b) Write a function to implement this operation using the linked stack implementation